

PARTS LIST

Semiconductors:

- IC1 - NE5534 op-amp
 D1-D6 - 1N4148 signal diode
 T1 - TIP141 npn transistor
 T2 - TIP146 pnp transistor
 ZD1-ZD4 - 10V, 1.3W Zener diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon, unless stated otherwise):

- R1 - 100-ohm, 1W
 R2, R3 - 200-ohm, 1W
 R4, R6 - 1-kilo-ohm
 R5 - 100-kilo-ohm
 R7 - 30-kilo-ohm
 R8, R10 - 100-ohm
 R9 - 150-ohm
 R11, R12 - 0.15-ohm, 2W
 R13 - 1-ohm, 5W

Capacitors:

- C1 - 0.1 μ F, 630V polyester
 C2-C5 - 0.1 μ F ceramic disk
 C6-C9 - 4700 μ F, 40V electrolytic
 C10, C11 - 220 μ F, 40V electrolytic
 C12 - 1 μ F, 40V electrolytic
 C13 - 100pF ceramic disk
 C14, C15, C17, C19 - 0.33 μ F ceramic disk
 C16, C20 - 1000 μ F, 40V electrolytic
 C18 - 0.15 μ F ceramic disk

Miscellaneous:

- S1 - On/off toggle switch
 CON1 - 2-pin terminal connector
 CON2, CON3 - 5-pin connector
 CON4, CON5 - 2-pin connector
 F1 - 0.5A fuse
 BR1 - 3A bridge rectifier
 LS1 - 4-ohm, 30W speaker
 X1 - 230V AC primary to 22V-0-22V, 3A secondary transformer
 Heat-sinks for transistors
 8-pin base/socket for IC1

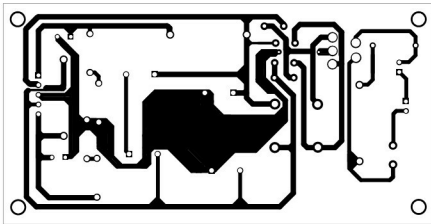


Fig. 3: Actual-size PCB layout of the power supply

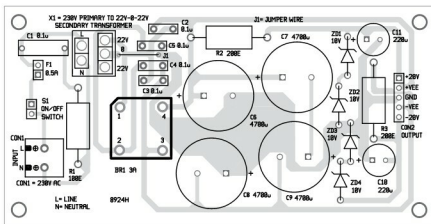


Fig. 4: Components layout for the PCB in Fig. 3

Output current. The output current should be limited to around 4A. Diodes D1 through D3 and resistor R11 limit the output current of T1 (npn). Diodes D4 through D6 and resistor R12 limit the output current of T2 (pnp). Diodes D1 through D6 (each 1N400X or 1N4148) are mounted on the heat-sinks of transistors T1 and T2.

Here, $R11 = R12 = R = 0.15\text{-ohm}$ to 0.22-ohm. The maximal output current is calculated as:

$$I_{\text{max}} = 0.65V/R = 0.65V/0.18\text{-ohm} = 3.6A \text{ or}$$

$$I_{\text{max}} = 0.65V/R = 0.65V/0.15\text{-ohm} = 4.3A$$

Output power. The output power of the amplifier depends on the power supply, power transistors, the size of heat-sinks and the load resistance. The heat-sink should have thermal resistance below 2°C/W and transistors should be isolated from

the heat-sink. The amplifier outputs around 30W, which is enough to drive a 4-ohm load. Each of transistors T1 and T2 should have a gain of more than 800 at collector current of 4A.

The maximal power dissipation of each of the transistors should be preferably more than 40W and the peak output current more than 6A. Some of the suggested Darlington transistor pairs are listed in the table.

Power supply.

The power supply circuit is shown in Fig. 1. It outputs four voltages: +20V, -20V, +VEE and -VEE. Transformer X1 is rated at 22V-0-22V, 3A. Bridge rectifier BR1 can be formed using 1N540X, CBR3A or BR305 diodes. Unregulated

SUGGESTED DARLINGTON POWER TRANSISTORS FOR THE AUDIO AMPLIFIER

npn	pnp	Ic	Ic(peak)	Vce(max)	Pd(max)
TIP141	TIP146	10A	15A	80V	125W
TIP142	TIP147	10A	15A	100V	125W
TIP121	TIP126	5A	8A	80V	65W
TIP122	TIP127	5A	8A	100V	65W
TIP101	TIP106	8A	15A	80V	80W
TIP102	TIP107	8A	15A	80V	80W
2N6301	2N6299	8A	16A	80V	75W
BD647	BD648	8A	12A	80V	62.5W
BD649	BD650	8A	12A	100V	62.5W
BD651	BD652	8A	12A	120V	62.5W

power supply ($\pm VEE$) is used only for the output Darlington transistors. It can go up to $\pm 40V$ depending on the transistors. The regulated power supply is limited by resistors R2 and R3, and Zener diodes ZD1 through